Bach Pham

Richmond, IN | 765-259-2118 | bachpham.edu@gmail.com
2006coder.com | github.com/2006coder | linkedin.com/in/2006coder

Education

Earlham College | GPA: 3.98/4.00 05/2028
Third-year B.A. in Computer Science, Honors Program | Minors: Mathematics and Physics
A+: AI & Machine Learning, Algorithms, Data Structures, Linear Algebra, Principles of Computer Organization
A: Quantum Mechanics, Modern Physics, Multivariate Calculus (Calc 3)

Research Experience

LANCE-7R: Multitask Enzyme Kinetics Prediction | First author Summer 2026-Present
Laboratory of Molecular and Thermodynamic Modeling | University of Maryland

- **Outperformed retrained CatPred** (**Nature Communications**, 2025; 140+ citations) in **18/18 Heldout/OOD99 metric point estimates** and **41/48** across all supplied cohorts; raised turnover OOD99 R^2 from **0.323 to 0.395** on 348 matched rows.
- Combined frozen **600M-parameter ESM++ protein language model** features, **RDKit** graphs and a **directed message-passing neural network (D-MPNN)** in a 22.53M-parameter **Python/PyTorch** backbone using **CUDA BF16 mixed precision**.
- Coupled **early molecule-conditioned cross-attention** and shared latent actions with **2 Joint-Embedding Predictive Architecture (JEPA) objectives: binding and reaction**, **contrastive losses**, and **SIGReg-style regularization**.
- Predicts **3 kinetic parameters** (turnover, Michaelis and inhibition constants) **without products at inference** for enzyme characterization and inhibitor evaluation relevant to **drug discovery**; reduced held-out Ki MAE from **0.865 to 0.788** on **1,193 measurements**.
- Added **406-parameter label memory** with sequence-cluster exclusion. **Manuscript being revised for journal submission**; currently running **12 ablation experiments** (6 loss configurations, 2 paired seeds) via **Slurm** on UMD's Zaratan **high-performance computing (HPC)** cluster under `energybio-eng`, with 3 memory comparisons.

  Research mentors: Prof. Jeffery B. Klauda | jbklauda@umd.edu | 301-405-1320
  Kinjal Mondal | kinjal14@umd.edu | 240-936-3552

Research Intern | Perception and Robotics Group, University of Maryland Summer 2025

- Implemented image filters from scratch; explored COLMAP, Gaussian Splatting, and RLBench; derived the eight-point algorithm and SVD rank constraint for 3D reconstruction.
  Research mentors: Prof. Yiannis Aloimonos | jyaloimo@umd.edu
  Pavan Mantripragada, PhD | mppavan@umd.edu

Publications

3 published conference papers | 2 first-author papers

- **First author; presented at IEEE/ACM CHASE 2026.** A Two-Stage AI System for Interpretable ACL Tear Screening in Connected Health Environments. Achieved **0.848 AUC vs. 0.719 baseline** on 30 held-out patients. Engineered **3 custom MRI input channels** (intensity, Frangi vesselness, local variance), **learned slice attention**, and max pooling in a 2.5D ResNet50.
- **First author, IEEE ICSC 2025.** Word Definitions from Large Language Models. Compared generated definitions with **3 published dictionaries**; evaluated rare-word representations against GloVe and FastText.
- **Second author, IEEE ICSC 2025.** Prosody Analysis of Audiobooks. Studied **93 aligned book-audiobook pairs**; model predictions correlated better with narration than commercial TTS for pitch in **22/24 books** and volume in **23/24**.

Professional Experience

LibreChat Primary Maintainer | Earlham College Fall 2025-Present

- **Collaborate with Earlham ITS and CS teams** to build and operate an AI-powered platform serving **approximately 150 Earlham students daily**; integrated SSO access and self-service API keys for campus-hosted models.
- Migrated LLM inference from Ollama to **llama.cpp across 2 Linux nodes and 4 NVIDIA A5000 GPUs**; configured replica routing, failover, and GPU memory allocation to serve **models up to 80B parameters**.
- Implemented isolated coding workspaces with **15 GiB per-user quotas**, network restrictions, and read-only system files; built deployment backups and tested database recovery in isolated containers.

Student Systems Administrator | Earlham College Feb. 2025-Present

- Administer **Linux servers, scientific software and dependencies for all STEM departments**, including **CUDA-enabled PyTorch/TensorFlow Jupyter environments**; automated wildcard TLS renewal and restored shared home directories on **2 servers**.

Physics & Calculus Tutor | Earlham College Sept. 2026-Present

- Tutor **Electromagnetism, Waves and Optics (PHYS 230/235)** and **Calculus B** in 2 weekly tutoring hours; help students interpret equations and apply mathematical concepts to course problems.
- Previously served as **Calculus A teaching assistant** through May 2026, supporting students with problem-solving and coursework.

Selected Project

NYX-V Quadruped Robot | C++, ESP32, FreeRTOS
Built a **12-DOF** robot with a custom 3D-printed chassis and **dual-core** Wi-Fi and inverse kinematics control.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, Bash, SQL, LaTeX
AI & Data: PyTorch, TensorFlow, Hugging Face, RDKit, ESM++, scikit-learn, NumPy, pandas, OpenCV, COLMAP
Engineering: Linux, Git, Docker, CUDA, Slurm, React, llama.cpp, ESP32, inverse kinematics